Stavion Colquitt

Audio ML / AI Engineer · Recording Artist

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01 SKILLS

Machine learning

PyTorch · ONNX · NumPy · SciPy · scikit-learn · pandas · librosa · Whisper (ASR)

Audio & DSP

STFT · Mel spectrograms · MFCCs · Speech enhancement · Acoustic echo cancellation · Source separation · Pitch tracking (pYIN) · Feature extraction · Real-time audio · Dante / AES67 · Audio-over-IP

MLOps & systems

Docker · Git · Linux · GitHub Actions · MLflow · Airflow · MCP servers · REST APIs · Qdrant · SQLite · On-device / edge inference

Enterprise AV/UC

Cisco Webex · CUCM · SIP · QoS · Crestron · Shure DSP · Biamp DSP · Q-SYS · ServiceNow

02 EDUCATION

M.S. Artificial Intelligence

Lewis University · in progress, expected 2028

Focus: machine learning for audio: source separation, music information retrieval, neural audio processing. Coursework includes Machine Learning Operations (Airflow, MLflow) and Math for Data Science.

05 SUMMARY

Audio engineer and recording artist building machine learning for music and speech. I train and ship audio models (PyTorch, ONNX), maintain open-source source-separation tools. Day to day I run real-time audio in enterprise conferencing rooms, the same setting my speech-enhancement models are built for. Currently completing an M.S. in Artificial Intelligence.

06 EXPERIENCE

Senior AV/UC Engineer · EOS IT Solutions, Abbott Laboratories — June 2025 – present

- Primary point of contact for AV operations across Abbott's $650M multi-campus enterprise infrastructure; resolved 84% of the account's quarterly AV/UC tickets.
- Building an Abbott-approved AI platform that connects LLMs (Claude, Copilot, Gemini) to enterprise systems through MCP servers, with retrieval (RAG) over a data lake of operational data in SQLite, Qdrant and Chroma; Python and Node.js dashboards surface trends for leadership, deployed in Docker.
- Program Biamp Tesira, Q-SYS and Dante audio networks and the room DSP behind call audio, including acoustic echo cancellation (AEC) and noise reduction, and monitor latency, jitter and packet loss with ThousandEyes and Webex Analytics.
- Manage 300+ conference rooms through Crestron XiO, Webex Control Hub and XMS Cloud, and run AV for C-suite meetings and high-visibility events.
- Mentored an intern who went on to receive a full-time offer from the client.

AV/UC Systems Engineer · Unisys, Omnicom Media Group — Oct 2023 – June 2025

- Designed, deployed and maintained AV/UC for 150+ conference rooms and executive boardrooms at Omnicom's Chicago campus on Cisco, Crestron and Polycom, drafting room layouts in AutoCAD and Visio for new buildouts and retrofits.
- Programmed Dante audio networks, Shure Microflex systems, audio DSP (EQ, noise reduction) and Crestron control; administered Webex Control Hub for room management and analytics.

07 SELECTED PROJECTS

Audio ML Noise Reduction · PyTorch, ONNX, real-time DSP — **+4.9 dB SI-SDR · 16× real time**

- Built real-time background-noise suppression for video calls (Teams, Zoom) end to end: dataset, model, training and live pipeline, for the call audio I support every day across 300+ conference rooms.
- Custom U-Net trained on DNS Challenge 4, exported to ONNX for on-device, real-time CPU inference.
- v2 found and fixed a loss computed on log spectrograms instead of audio: +4.9 dB SI-SDR on the VoiceBank+DEMAND benchmark (v1: −5.0 dB), with training tracked in MLflow and the evaluation published in the repo.

B.S. Digital Audio Recording Arts
University of St. Francis, Joliet, IL · 2022

Recording Artist concentration: signal flow, acoustics, DSP, audio-over-IP and systems integration. Sang in the Choir and Jazz Choir.

03 CERTIFIED

CCNP Collaboration
Cisco · 2025 – 2028

Dante Level 3
Audinate · 2025 – 2028

04 MUSIC

Verified Spotify artist with released records on Spotify, Apple Music, YouTube and Pandora. I write, record, produce and mix my own music.
linktr.ee/stavionc

Nussl, modernized fork · Python, PyTorch, librosa — Open source

- Revived the Interactive Audio Lab's source-separation library, archived in 2024, so it installs and runs on current Python, NumPy, SciPy, librosa and PyTorch.
- Added LeadBacking, a separator that splits lead from backing vocals using a stereo panning index and pYIN pitch tracking, with uncertainty flags and documented failure cases.
- Fixed clean-install failures and added a fallback for the retired External File Zoo; tests run in GitHub Actions.

Audio Analysis MCP · Python, MCP, DSP

- MCP server giving AI assistants real-time audio analysis during a production session: frequency reports, stereo imaging, mix analysis and key detection.

Spectrogram Reader · Python, STFT

- Interactive STFT explorer with an A/B difference view for comparing two signals.

Hybrid RAG Pi · Docker, Node.js, Qdrant, SQLite

- Hybrid retrieval (Qdrant vectors + SQLite FTS5 keywords) with hierarchical conversation memory, running as 29 containerized services on a Raspberry Pi 5 (8 GB).

Thanks for listening. stavioncolquitt.com